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Closing the Divide: A Detailed Study on Gender Pay Disparity

In this study, we aim to analyze the gender pay gap across departments like Police, Health and Human Services, and Fire and Rescue Services, among others, within our organization. Our goal is to identify and address disparities in compensation between genders by examining salary data across roles, grades, and departments. Using the provided dataset, which includes detailed information on department codes, names, divisions, gender, base salary, overtime pay, longevity pay, and grade, we aim to uncover these disparities and propose solutions to promote fair and equitable compensation practices in the broader labor force.

Ensuring pay equity is crucial in fostering a workplace environment that values diversity and inclusion. It is essential to uphold principles of fairness and equality in all aspects of business operations. By addressing gender-based pay disparities, companies can strive to create an environment where every employee feels valued, respected, and fairly compensated for their contributions, regardless of gender. This is a significant step towards building a more inclusive and equitable workplace for all employees, contributing to the overall well-being and morale of the organization they are a part of.

The dataset provides detailed salary information for Montgomery County, MD, in 2023. It includes attributes such as department, gender, base salary, overtime pay, longevity pay, and grade. Departments are identified by code and name, with divisions specifying employee assignments. Gender is categorized as male or female. Base salary reflects annual projections based on employee status and hours. Overtime pay compensates for extra hours worked, while longevity pay rewards service length. Grades categorize employees by level. The dataset contains 10.3k valid entries, ranging from \$11.1k to \$292k in salary. Mean salaries are \$90.3k with a \$31.2k standard deviation. Overtime pay averages \$8.08k with a \$16.5k standard deviation, while longevity pay averages \$1.53k with a \$3.21k standard deviation.

The plotting functions for these metrics have been developed using R, with the following sequence comprising the code for the graph, the graph itself, and the caption.

```
ggplot(data, aes(x = Gender, y = Base_Salary, fill = Gender)) +  
  geom_boxplot() +  
  labs(title = "Gender-Based Salary Distribution",  
        x = "Gender",  
        y = "Base Salary in USD") +  
  scale_y_continuous(labels = scales::comma)+  
  theme_minimal()
```



Figure 1: Boxplot showing the distribution of base salaries by gender, with the x-axis representing base salary in USD and the y-axis representing gender.

Figure 1 shows that the average base salary for females is slightly lower than for males, but the difference is very small. The differences between the genders come into play when looking at the outliers. Males have significantly more outliers than females in the higher base salary direction.

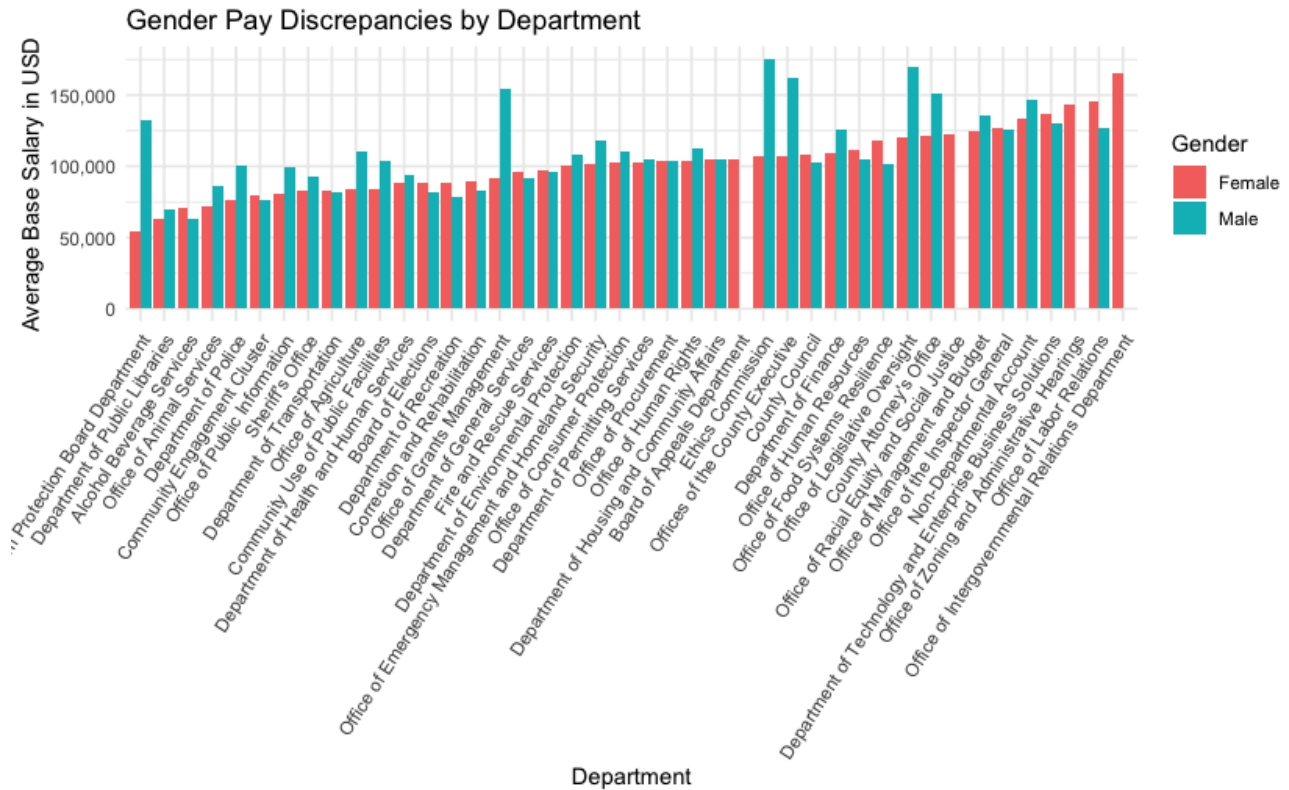
Code for Figure 2:

```
#calculate the average salary
average_salaries <- data %>%
  group_by(Department_Name, Gender) %>%
  summarise(Average_Base_Salary = mean(Base_Salary, na.rm = TRUE)) %>%
  spread(Gender, Average_Base_Salary) %>%
  arrange(F) # Arrange by female average base salary in ascending order

# Rename columns to make them easier to work with.
colnames(average_salaries)[colnames(average_salaries) == "F"] <- "Female"
colnames(average_salaries)[colnames(average_salaries) == "M"] <- "Male"

# Reshape for plotting.
department_order <-
average_salaries$Department_Name[order(average_salaries$Female)]

# Plot using the ordered departments
ggplot(long_data, aes(x = factor(Department_Name, levels = department_order), y
= Salary, fill = Gender)) +
  geom_bar(stat = "identity", position = "dodge") +
  labs(title = "Gender Pay Discrepancies by Department",
        x = "Department",
        y = "Average Base Salary in USD",
        fill = "Gender") +
  scale_y_continuous(labels = scales::comma) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 57, hjust = 1))
```



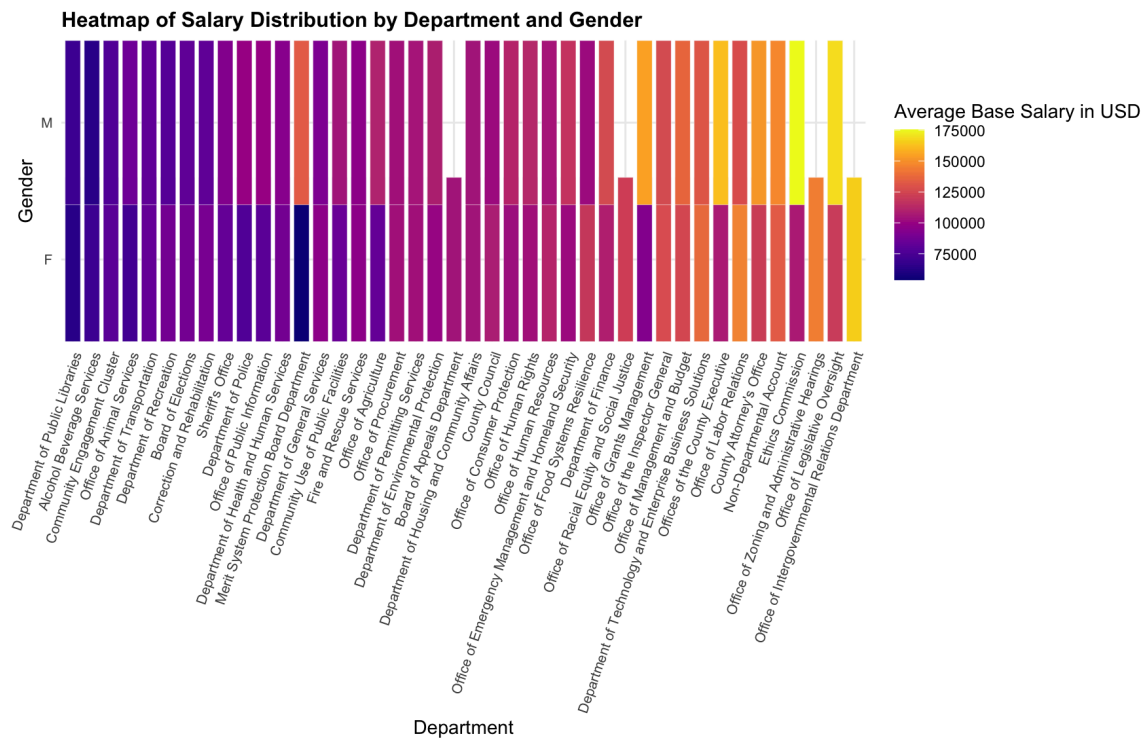
Note: In some categories there are only females that work in the department, so there is only a female bar with a white space next to it.

Figure 2: Bar chart illustrating gender pay discrepancies by department, with the x-axis representing department names and the y-axis indicating the average base salary in USD. The bars are grouped by gender, showing the mean salary for each department. The male salaries are on the right, while the female salaries are on the left. The female salaries are ordered by lowest to highest, showing a channel.

From Figure 2 it can be seen there are some departments with large spikes dominated by males, but then for the other departments, the disparity is less pronounced. Even with the disparity less pronounced, most of the male average base salaries are higher than their female counterparts.

Code for Figure 3:

```
ggplot(avg_salary, aes(x = Department_Name, y = Gender, fill = Base_Salary)) +
  geom_tile(color = "white", width = 0.8, height = 1.2) +
  scale_fill_viridis(name = "Average Base Salary in USD", option = "plasma") +
  labs(title = "Heatmap of Salary Distribution by Department and Gender",
       x = "Department",
       y = "Gender") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 71, hjust = 1, size = 8),
        axis.text.y = element_text(size = 8),
        plot.title = element_text(size = 12, face = "bold"))
```



Note: The color represents the range of average base salary in USD and cannot vary between different colors, as this would not effectively illustrate the differences in average base salary between females and males.

Figure 3: Heatmap illustrating the average base salary distribution by department and gender. The color gradient represents the average base salary in USD, with warmer colors indicating higher salaries. Departments are listed on the x-axis, while genders are displayed on the y-axis. The plot provides a visual overview of salary distribution across different departments and genders within the organization. The heatmap is organized by gender with male salaries being represented in the top half, and female salaries in the bottom half.

In Figure 3, there are a couple of departments that stand out in terms of pay disparity. Merit System Protection Board Department, Office of Grants Management, Offices of the County Executive, and Ethics Commission all can be seen to have significant differences in pay between males and females, where males earn more than females. There are examples of the reverse as well in the Office of Food Systems Resilience and the Office of Labor Relations, but the disparities between those departments are less than the differences found in the male-favored departments.

Code for Figure 4:

```
avg_salary_by_grade$Grade <- as.numeric(as.factor(avg_salary_by_grade$Grade))
correlation <- cor(avg_salary_by_grade$Grade, avg_salary_by_grade$avg_salary)
ggplot(avg_salary_by_grade, aes(x = Grade, y = avg_salary)) +
  geom_point() +
  labs(title = "Correlation between Grade and Average Salary",
        x = "Grade",
        y = "Average Salary in USD") +
  geom_smooth(method = "lm", se = FALSE) + # Add a linear regression line
  annotate("text", x = 0.5, y = max(avg_salary_by_grade$avg_salary),
          label = paste("Correlation:", round(correlation, 2)), size = 3,
  color = "red") +
  scale_y_continuous(labels = scales::comma)+
  theme_minimal()
```

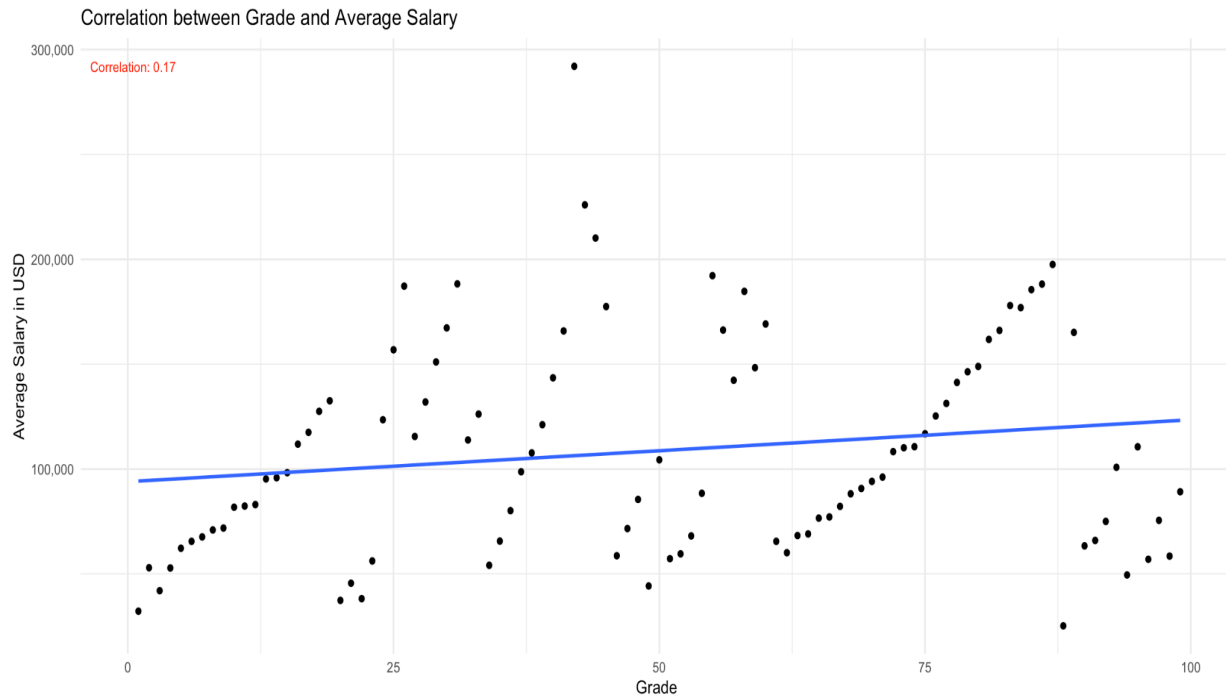


Figure 4: Scatter plot displaying the correlation between employee grade (Grades categorize employees by level) and average salary in USD. Each point represents the average salary corresponding to a specific employee level. A linear regression line is added to visualize the trend in the data. The correlation coefficient is indicated in red text, providing insight into the strength and direction of the relationship between grade and salary.

Figure 4 illustrates that there is a general trend of increasing average salaries with ascending employee levels. However, the dispersion of individual salaries within these levels reveals a more complex narrative. The data demonstrates that a lower employee level does not invariably correspond to a lesser salary, nor does a higher level guarantee a greater one. This observation suggests that while pay grades are intended to reflect experience and responsibility, they are not the sole determinants of salary levels. Importantly, the variance observed within the same grade levels indicates that there may be additional variables at play that influence earnings beyond the defined grade structure. It is within this variability that we must examine the subtleties of gender pay disparity, recognizing that grade level alone does not account for the nuanced differences in compensation between males and females.

Code for Figures 5/6/7:

```

dept_data <- data[grep("Department", data$Department_Name), ]

# Check if dept_data is empty
if (nrow(dept_data) == 0) {
  print("No entries found with 'Department' in the Department_Name column.")
} else {
  # Select relevant columns
  salary_attributes <- dept_data[, c("Base_Salary", "Overtime_Pay",
"Longevity_Pay")]
  salary_attributes$Group <- dept_data$Department_Name

  ggparcoord(salary_attributes,
              columns = c("Base_Salary", "Overtime_Pay", "Longevity_Pay"),
              groupColumn = "Group",
              scale = "std", # Scale the data based on standard deviation
              alphaLines = 0.5,
              order = "allClass") + # Order by class
  labs(title = "Parallel Coordinates Plot of Salary Attributes",
       x = "Salary Attributes in USD",
       y = "Value") +
  theme_minimal() +
  theme(plot.title = element_text(size = 16, face = "bold"),
        axis.title.x = element_text(size = 14),
        axis.title.y = element_text(size = 14),
        axis.text = element_text(size = 12))
}

office_data <- data[grep("Office", data$Department_Name), ]

# Check if office_data is empty
if (nrow(office_data) == 0) {
  print("No entries found with 'Office'")
} else {
  salary_attributes <- office_data[, c("Base_Salary", "Overtime_Pay",
"Longevity_Pay")]
  salary_attributes$Group <- office_data$Department_Name

  ggparcoord(salary_attributes,

```



```

        columns = c("Base_Salary", "Overtime_Pay", "Longevity_Pay"),
        groupColumn = "Group",
        scale = "std", # Scale the data based on standard deviation
        alphaLines = 0.5,
        order = "allClass") + # Order by class
labs(title = "Parallel Coordinates Plot of Salary Attributes",
      x = "Salary Attributes in USD",
      y = "Value") +
theme_minimal() +
theme(plot.title = element_text(size = 16, face = "bold"),
      axis.title.x = element_text(size = 14),
      axis.title.y = element_text(size = 14),
      axis.text = element_text(size = 12))
}

no_dept_data <- data[!grepl("Department", data$Department_Name), ]
leftover_data <- no_dept_data[!grepl("Office", no_dept_data$Department_Name), ]

# Check if office_data is empty
if (nrow(leftover_data) == 0) {
  print("No entries found without 'Department' or 'Office'")
} else {
  salary_attributes <- leftover_data[, c("Base_Salary", "Overtime_Pay",
"Longevity_Pay")]
  salary_attributes$Group <- leftover_data$Department_Name

ggparcoord(salary_attributes,
            columns = c("Base_Salary", "Overtime_Pay", "Longevity_Pay"),
            groupColumn = "Group",
            scale = "std", # Scale the data based on standard deviation
            alphaLines = 0.5,
            order = "allClass") + # Order by class
labs(title = "Parallel Coordinates Plot of Salary Attributes",
      x = "Salary Attributes in USD",
      y = "Value") +
theme_minimal() +
theme(plot.title = element_text(size = 16, face = "bold"),
      axis.title.x = element_text(size = 14),

```

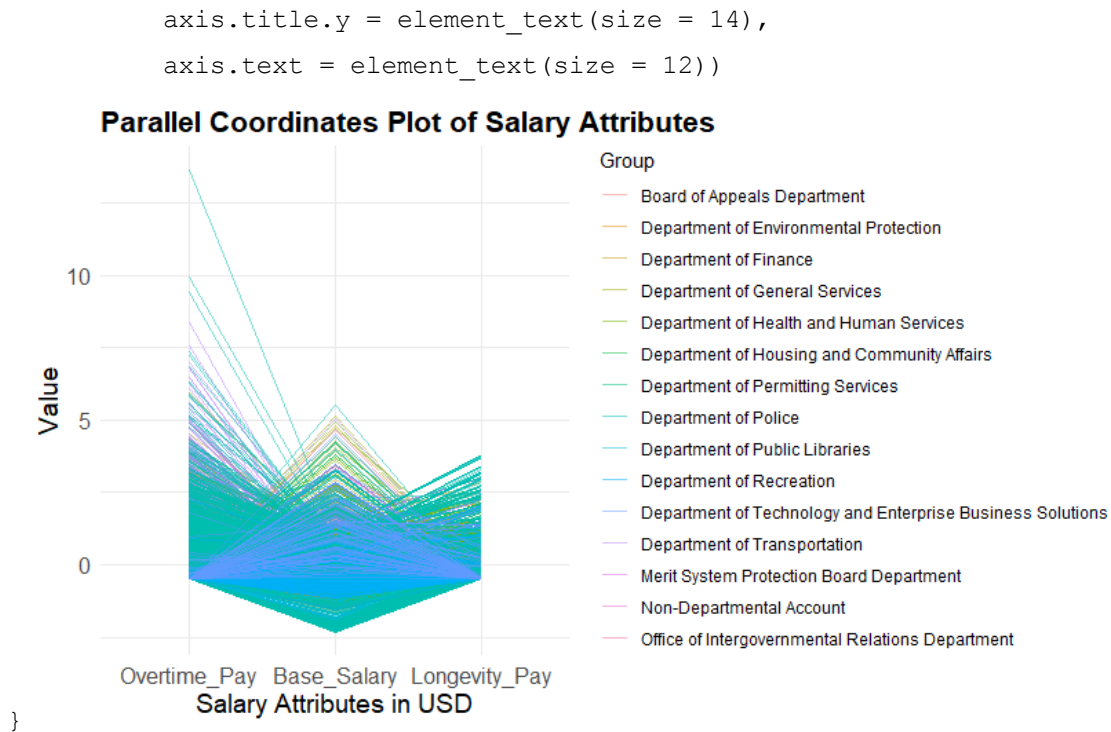


Figure 5: Parallel coordinates plot illustrating the distribution of salary attributes (among only the departments in the data set) including base salary, overtime pay, and longevity pay across different departments. Each line represents a department, with variations in salary components indicated by the position along the vertical axes. The plot is scaled based on the standard deviation for comparison. The title emphasizes the visualization's focus on salary attributes, while axis labels provide clarity on the represented variables.

Figure 5 shows two dominant shapes with a very tight spread. One shape can be seen as a triangle, where departments in this group have no overtime pay or longevity pay, but have a higher base salary. The second shape can be seen as a V shape, where departments in this group are paid overtime and longevity pay, but have a lower base salary as compared to the triangle department group. This most likely comes from the distinction between hourly and salaried employees. This is because salaried employees tend to have a higher base salary than hourly employees, but are not offered overtime pay.

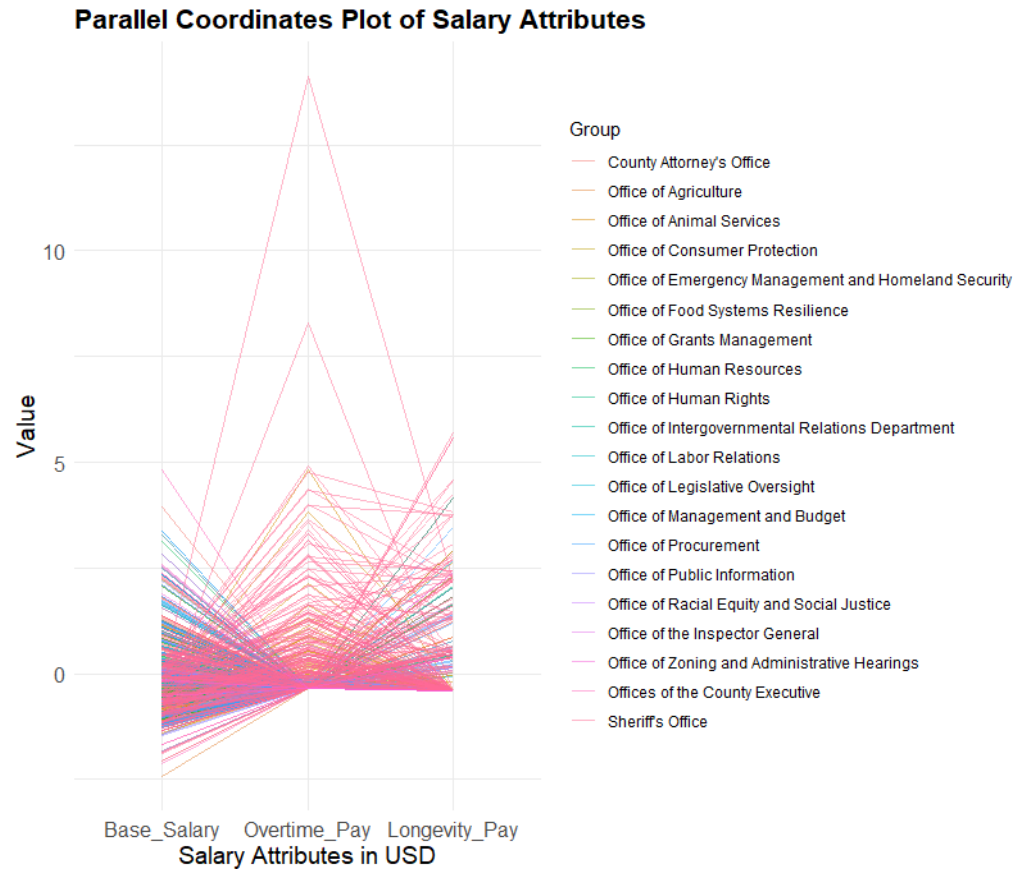


Figure 6: Similar plot to figures 5 and 7, but only looks at the Offices represented in the data set.

Figure 6 has a much larger spread than the one shown in Figure 5. We see a similar shape represented across the whole plot, but it seems like the base salaries are lower overall. Aside from a couple outliers, there seems to be similar overtime pay to Figure 5. This plot has these differences likely because of the differences in funding between offices and departments. The whole graph plot looks scaled down when compared to Figure 7. It is hard to say why that is the case here due to the variation of positions in Figure 7.

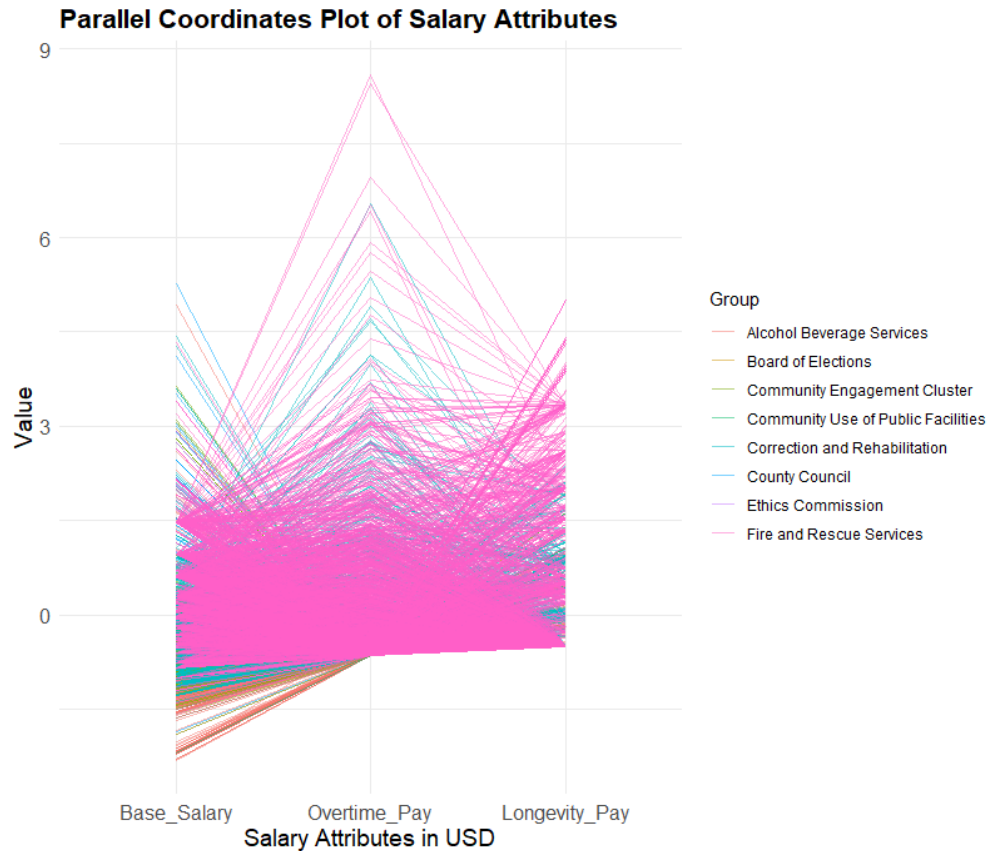


Figure 6: Similar plot to figures 5 and 6, but looks at the group of positions not categorized by “Department” or “Office”.

Figure 7 has a much larger spread across all 3 types of pay. This is likely caused by the lack of correlation in position. Figure 7 displays the largest variation among the 3 figures being compared. It holds the same base shape as Figures 5 and 7, but the shapes here are much harder to see. It also seems that there is a much larger representation in the Fire and Rescue Services than any other group in this figure. This likely means that there is a lack of representation in the other categories represented here in comparison. That being said, we can see a large variation in pay among all 3 pay categories even when just looking at the Fire and Rescue Services.

References:

Dataset-

<https://www.kaggle.com/datasets/sahirmaharajj/employee-salaries-analysis?resource=download>